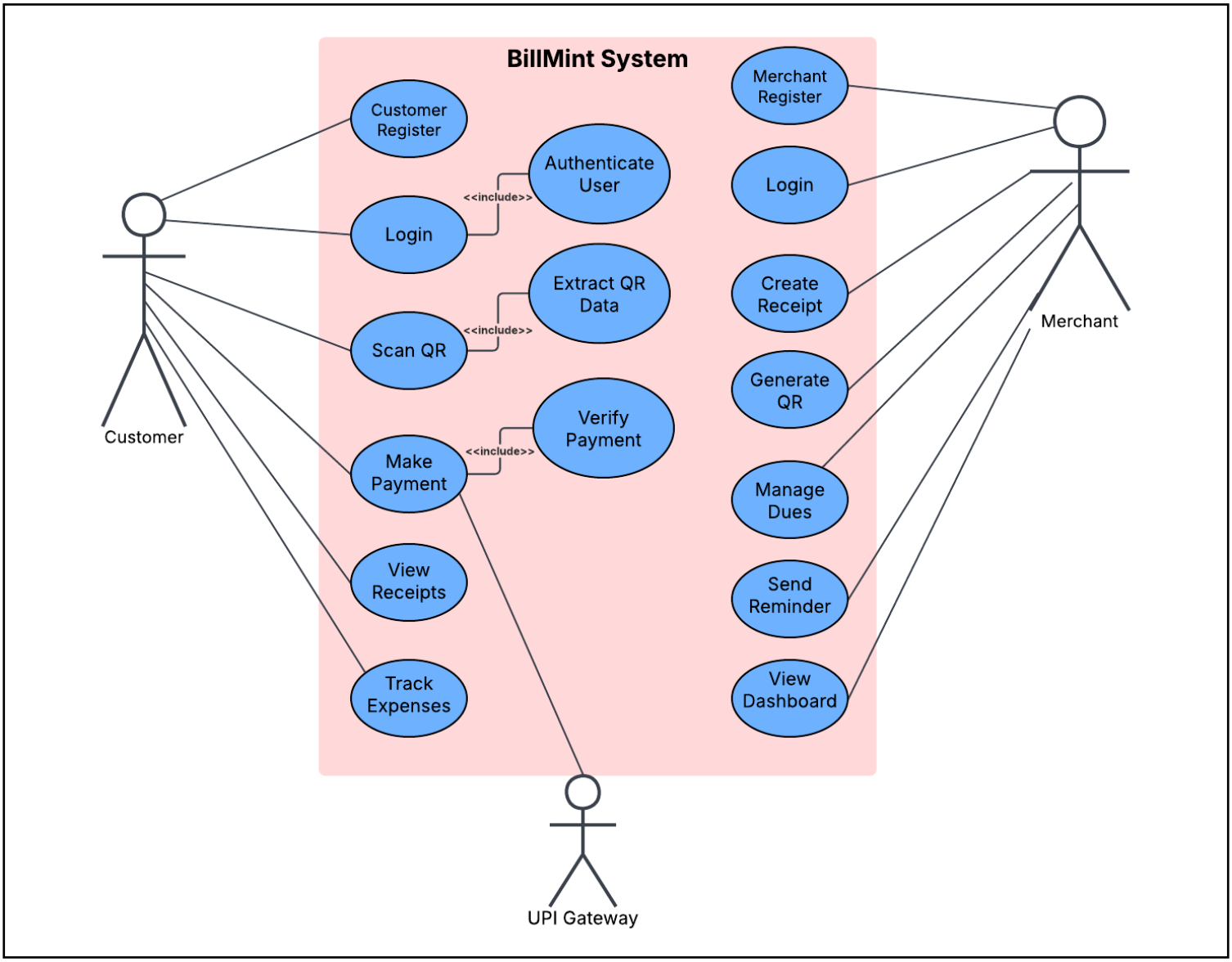
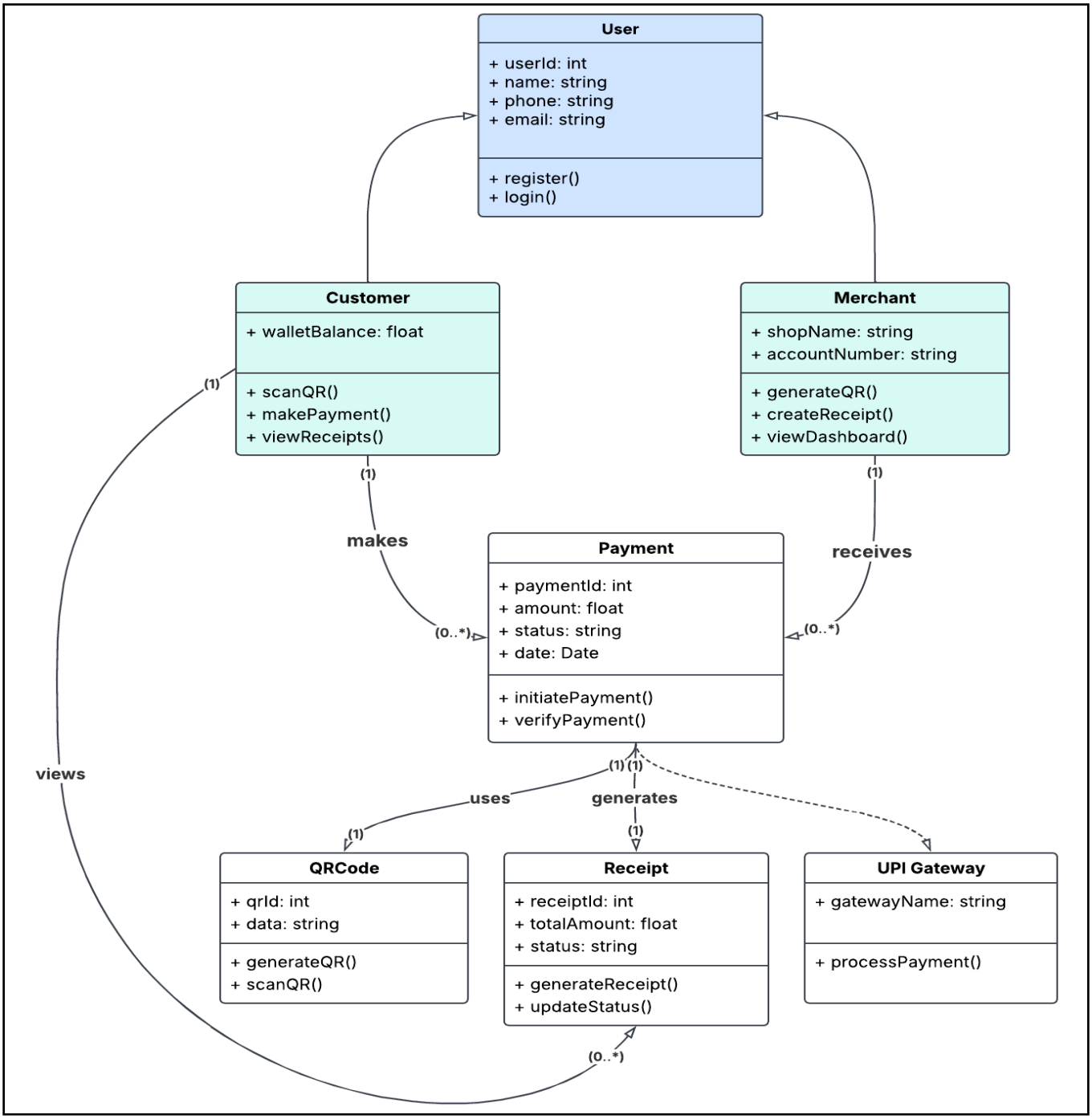
Use Case Diagram of BillMint System



The Use Case Diagram represents the functional interactions between external actors and the BillMint system. It identifies three primary actors: Customer, Merchant, and UPI Gateway. The Customer can perform actions such as registration, login, scanning QR codes, making payments, viewing receipts, and tracking expenses. The Merchant is responsible for generating QR codes, creating receipts, managing dues, sending reminders, and viewing dashboards. The system also interacts with the UPI Gateway to process and verify payments.

The diagram highlights important relationships such as «include», where functionalities like authentication, QR data extraction, and payment verification are reused within other use cases. This diagram provides a high-level overview of system functionalities and clearly defines the responsibilities of each actor. It helps in understanding the system scope and serves as a foundation for further design and development.

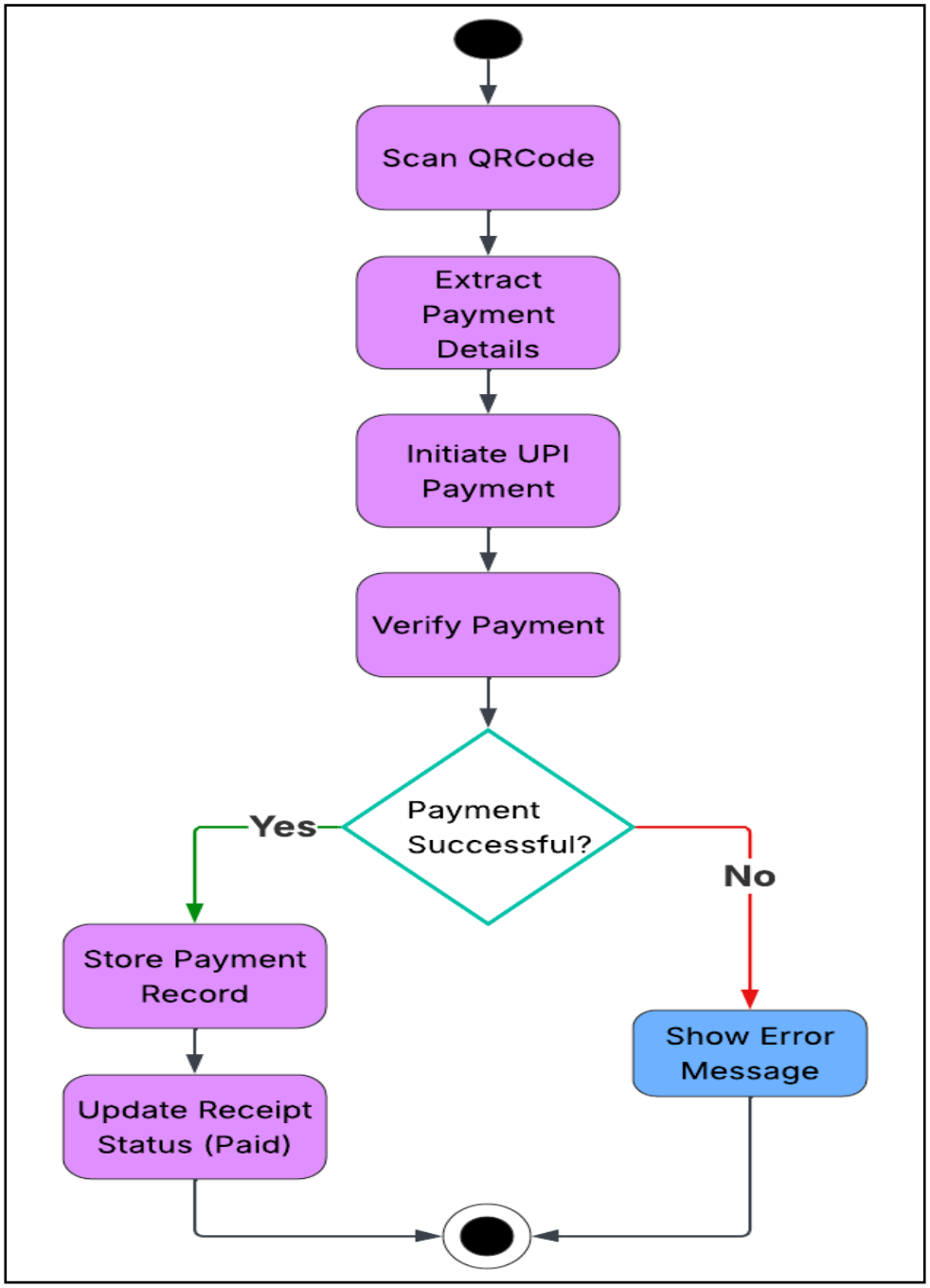
Class Diagram of BillMint System



The Class Diagram illustrates the structural design of the BillMint system, showing the main classes, their attributes, methods, and relationships. The *User* class is generalised into *Customer* and *Merchant*, representing inheritance. The *Customer* performs actions like scanning QR codes and making payments, while the *Merchant* generates QR codes and manages receipts.

The *Payment* class acts as a central entity connecting customers and merchants. Additional classes such as *QRCode*, *Receipt*, and *UPI Gateway* support QR processing, receipt generation, and payment handling. This diagram provides a clear view of the system's architecture and interactions between different components.

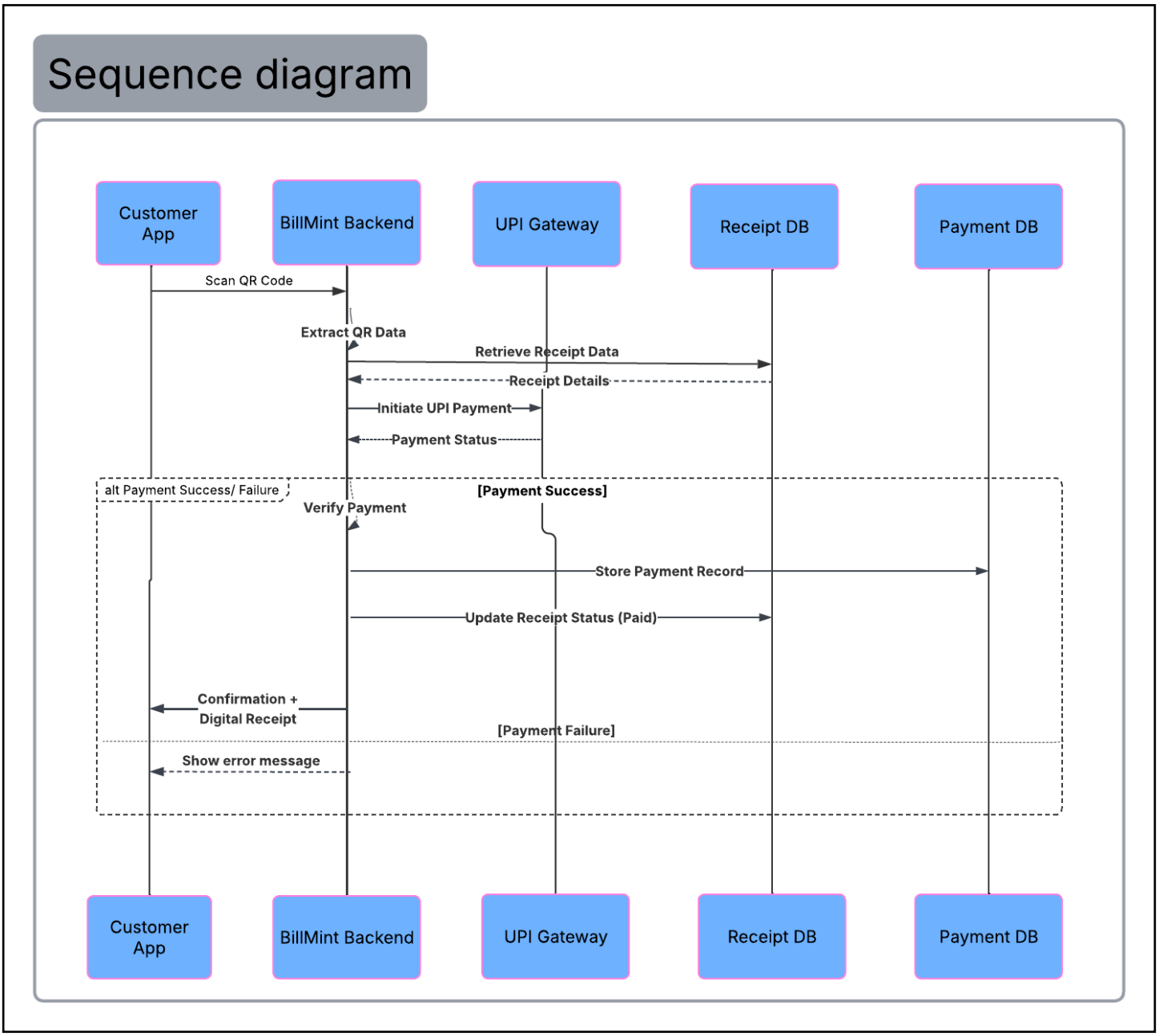
Activity Diagram of BillMint System



The Activity Diagram represents the workflow of the payment process in the BillMint system. It begins with scanning a QR code, followed by extracting payment details and initiating the UPI payment. The system then verifies the payment status received from the gateway.

A decision node checks whether the payment is successful or not. If successful, the system stores the payment record and updates the receipt status as paid. If the payment fails, an error message is displayed to the user. The process then ends after completing the respective flow. This diagram helps in understanding the step-by-step execution of system activities.

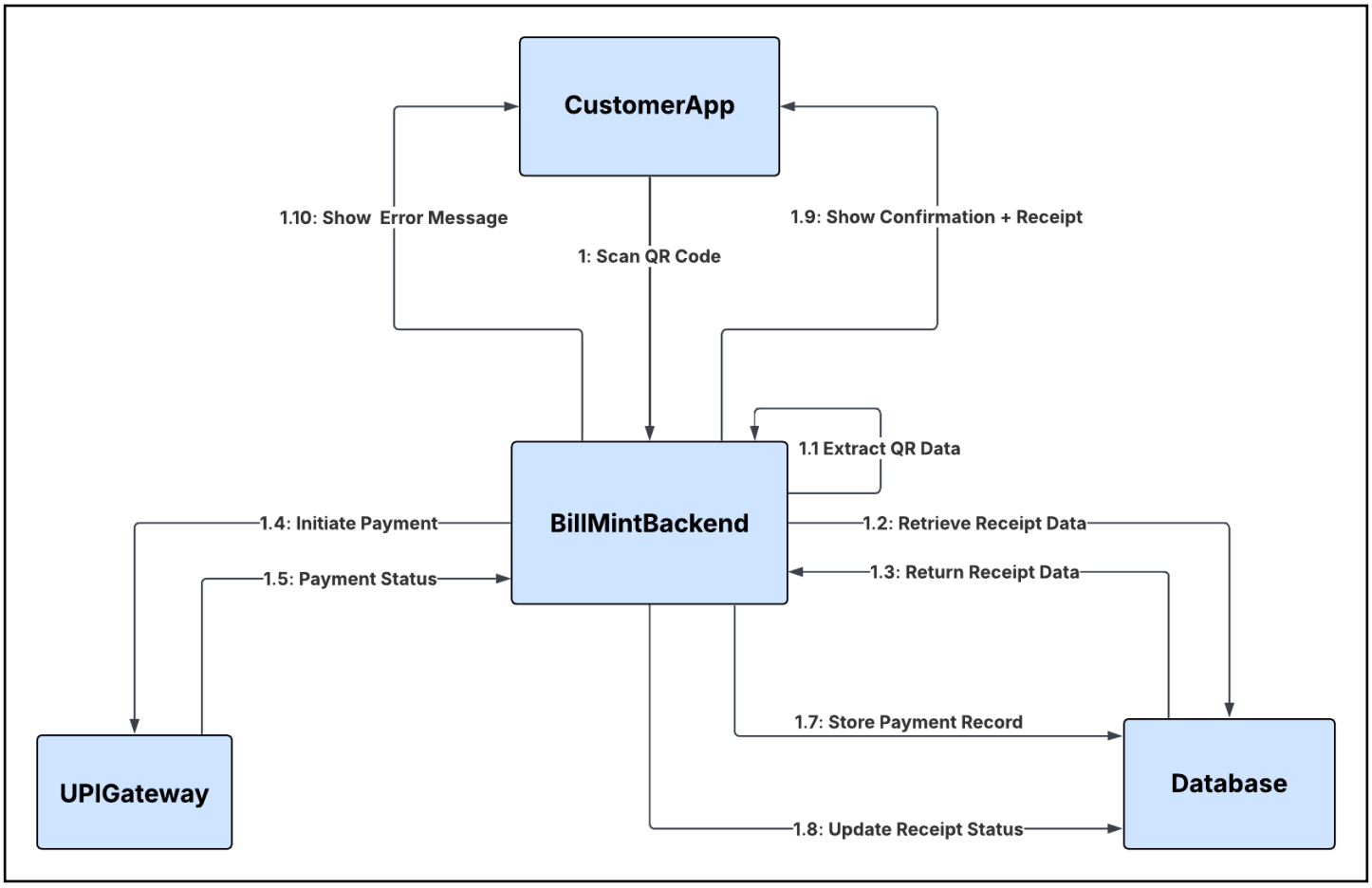
Sequence Diagram of BillMint System



The Sequence Diagram illustrates the step-by-step interaction between system components during a payment process. The main entities involved are *Customer App*, *BillMint Backend*, *UPI Gateway*, *Receipt Database*, and *Payment Database*. The process begins with the customer scanning a QR code, after which the backend extracts the data and retrieves receipt details from the database.

The backend then initiates the payment through the UPI Gateway and receives the payment status. Based on the result, the system either stores the payment record and updates the receipt status or displays an error message to the user. The use of an alternative (alt) fragment represents both success and failure scenarios. This diagram helps in understanding the chronological flow of operations in the system.

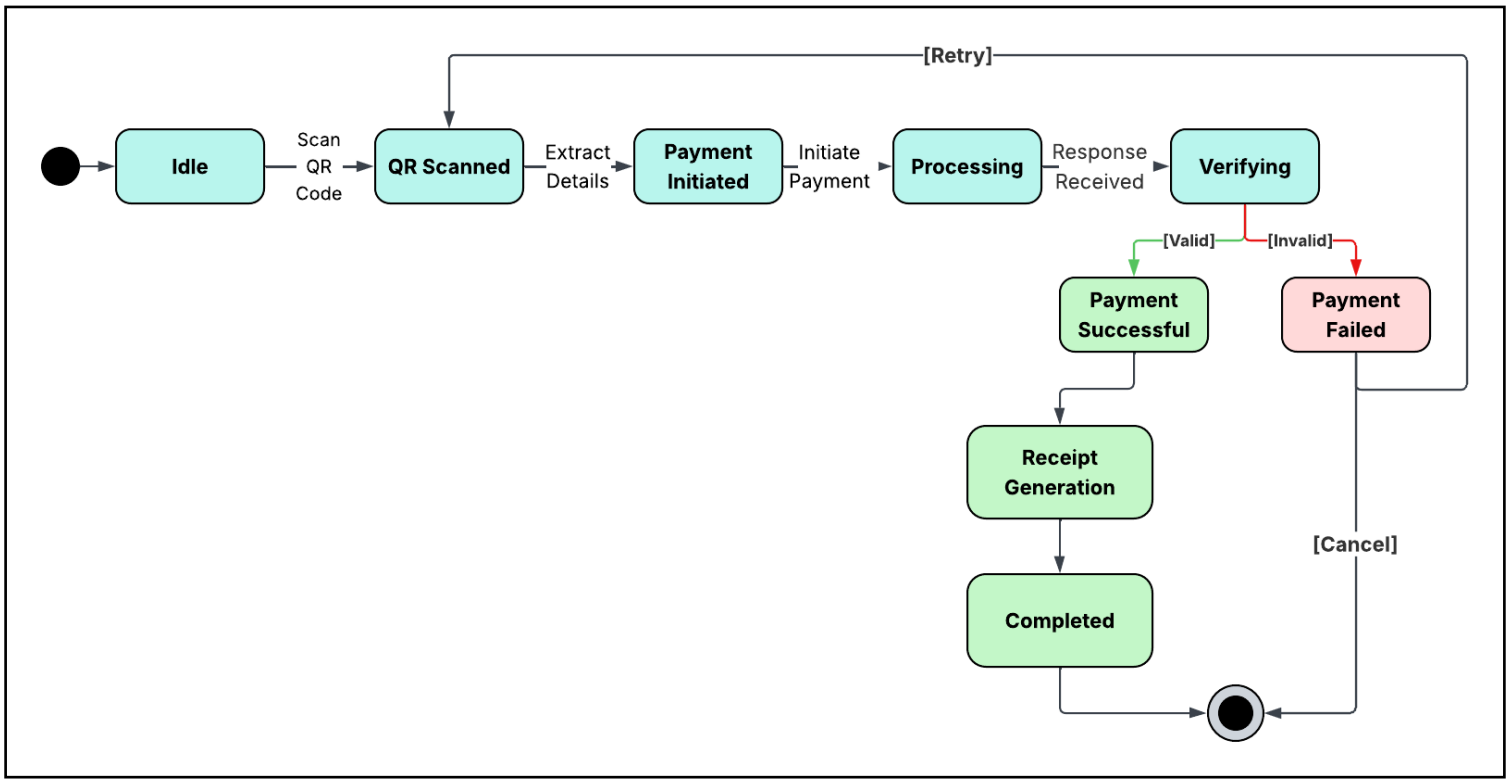
Collaboration Diagram of BillMint System



The Collaboration Diagram illustrates the interaction between different system components to complete a payment process. The main objects involved are *CustomerApp*, *BillMintBackend*, *UPI Gateway*, and *Database*. The diagram shows how the customer scans a QR code, after which the backend processes the request by extracting data and retrieving receipt information from the database.

The backend then initiates the payment through the UPI Gateway and receives the payment status. Based on the result, it either stores the payment record and updates the receipt status or displays an error message to the user. The numbered messages indicate the sequence of interactions between objects. This diagram helps in understanding how different components collaborate to execute the system functionality.

State Diagram of BillMint System



The State Diagram represents the different states of the payment process in the BillMint system and the transitions between them. It begins with the *Idle* state, followed by *QR Scanned*, *Payment Initiated*, and *Processing* states as the user scans a QR code and initiates a payment. The system then moves to the *Verifying* state after receiving a response from the UPI Gateway.

Based on the validation result, the system transitions to either *Payment Successful* or *Payment Failed*. In case of success, a receipt is generated and the process ends in the *Completed* state. In case of failure, the user can either retry the process or cancel it. This diagram helps in understanding the dynamic behaviour of the system during the payment lifecycle.